

1 **GCerrHMM: a GC-aware error hidden Markov model for**
2 **long-read sequencing simulation with a reproducible**
3 **fidelity harness**

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7 Article type: Research article | Target journal: BMC Bioinformatics | Collection "Simulated
8 and synthetic data in bioinformatics: statistical methods, models, and applications" | W3.3
9 submission-format manuscript, 2026-09-28.

10

11 **Abstract**

12 **Background**

13 Accurate simulation of long sequencing reads is a prerequisite for benchmarking alignment,
14 variant-calling and assembly pipelines when ground truth is unavailable. Existing simulators
15 model sequencing error as a global process, yet real instruments generate error whose rate and
16 structure vary with local sequence context, including GC content. A simulator that claims to
17 capture such context dependence needs an evaluation instrument that can actually measure it.

18 **Results**

19 We present GCerrHMM, a transition-based error HMM trained on real ONT alignments
20 whose state-transition model is conditioned on local GC content, and a reproducible fidelity
21 harness. Across six genomes the GC-aware model was directionally closer to the real GC-error
22 curve than its one-bin control in both replicates for four species, including the GC-heterogeneous
23 *H. sapiens* chr21; *A. thaliana* and *D. melanogaster* were mixed. We report effect sizes and the
24 precision of the between-window comparison at 10x, 20x and 30x. In a separate depth grid the
25 read-sampling SD ranged from 0.009 to 0.194 across 5-15x (6/3/2 subsets); the 15x estimate is
26 suggestive rather than precise. The composite distributional score, which contains no GC-
27 conditional term, is reported as a descriptive summary; the GC-stratified instrument carries the
28 GC-specific comparison. On the coverage-matched chr21 30x panel, GCerrHMM ranked second
29 on alignment identity and first on assembly Borda distance, meeting the pre-registered top-2
30 criterion in both anchored layers.

31 **Conclusions**

32 GCerrHMM provides a trainable GC-aware error model and a reproducible evaluation
33 harness. Its GC-conditional effect is measurable in structure; we report the effect size, the
34 separate sampling and structural uncertainties, and the depth needed for a definitive cross-species
35 test. The study also shows that simulator benchmarking requires instrument-claim alignment: a

fidelity score can test only the property it contains.

Keywords: long-read sequencing; read simulator; hidden Markov model; GC bias; Oxford Nanopore; benchmark; reproducibility

Background

Long-read sequencing platforms (Oxford Nanopore Technologies, ONT; Pacific Biosciences HiFi) have become the default for de novo assembly, structural-variant discovery and haplotype phasing. Every methodological advance on these platforms is benchmarked, and benchmarking requires either real data with known truth - rare and expensive - or simulated reads whose error process reproduces the instrument. The fidelity of the simulator therefore bounds the validity of every conclusion drawn from the benchmark.

Widely used long-read simulators, including NanoSim [1] and PBSim3 [2], estimate error or quality profiles from aligned reads and then apply those profiles during simulation. Empirical simulators such as NPBSS [3] likewise prioritise realistic read-level error and quality distributions. These approaches provide strong baselines, but their primary model descriptions do not expose a local-GC-conditioned transition model. The evaluation literature also shows that simulator rankings depend on the metric: distributional, alignment, variant, and assembly endpoints can favour different tools [1-3]. Recent work has expanded the field toward metagenomic profiling [4], tunable signal-level generation [5, 6], transformer-based signal models [7, 8], context-aware simulation for parameter optimisation [9], and more rigorous long-read benchmark references [10, 11]. A 2026 preprint benchmark of six ONT read simulators reported that no tool reproduced all read-level and error-structure metrics, and specifically found that PBSim3 did not capture context-dependent substitution rates or homopolymer-length errors [11]. GC conditioning therefore requires both a model mechanism and a measurement instrument aligned to that mechanism. The GC context of the six-genome panel and representative GC-error curves are shown in Fig. 1.

A simulator that conditions its error model on local GC content is straightforward to propose; the harder problem is measurement. If the evaluation instrument cannot see GC-conditional error, then no experiment can distinguish a GC-aware model from a GC-unaware one, and the literature gains an untestable claim. We therefore treat the evaluation harness as a first-class contribution alongside the model itself: every claim in this paper names the instrument it rests on. Fig. 2 summarizes recent simulator designs and the modelling axes they cover.

Here, we present GCerrHMM, an error-state hidden Markov model trained from real long-read alignments whose transition structure is conditioned on local GC content, with a length-aware indel encoding that removes a structural bias we show is present in single-state encodings. We evaluate it with a two-level harness (distributional fidelity; downstream task consistency) across six genomes and four simulators, with pre-registered decision rules for the central GC claim. We report the observed directional effects, the precision limits of the current instrument, and the depth budget required for a definitive cross-species test. Fig. 3 places the present study within that context and evaluation chain.

75

76 **Results**

77 **A GC-aware error model trained from real alignments**

78 GCerrHMM is an error-state HMM trained from primary alignments of real long reads (Fig.
79 4a). The reference sequence is tiled into 100-bp windows, each window is assigned to one of K
80 equal-width GC bins, and transition probabilities $P(\text{state at } i+1 \mid \text{state at } i, \text{GC bin})$ are estimated
81 by counting along the CIGAR path, with hierarchical smoothing of per-bin counts toward a
82 global prior. The default state space has ten states - M, S, I1, I2, I3, I4+, D1, D2, D3, D4+ - in
83 which indel run length is carried in the state name; a simplified state space merging indels into
84 single I and D states is used as an ablation control. The same training chain supports ONT and
85 HiFi data, and the HiFi chain completes end to end (Fig. 4a).

86 The model-implied error rate per GC bin recovers the qualitative shape of the real data
87 where such structure exists: in *A. thaliana*, whose genome carries strong local GC heterogeneity,
88 the trained model reproduces a sharp error peak in the highest-GC bin (19.5% in bin 7), while
89 the curves for the GC-uniform *E. coli* and *S. cerevisiae* genomes are flat (Fig. 4b). Fig. 4b is
90 model-implied; the real-versus-simulated GC-error curves used to motivate the model are shown
91 in Fig. 1b. The six-genome panel and its data sources are summarised in Fig. 4c.

92 Fig. 4c lists the six analyzed references, RefSeq assemblies, genome sizes, GC content, GC
93 heterogeneity and ONT run accessions. Table 1 provides the same source panel in tabular form.

94 ***Length-aware indel encoding removes a structural insertion-length cap***

95 With a single insertion state, a one-base and a fifty-base insertion collapse to the same state,
96 the I-to-I transition is never observed, and the learned model emits every insertion as exactly 1
97 bp. On the human chr21 training alignment the mean insertion length is 3.83 bp, and the single-
98 state encoding produces 0.04% inserted bases against 3.78% under the same CIGAR-derived M-
99 base denominator. The run-length encoding (I1-I3, I4+; D1-D3, D4+, with a heavy-tailed length

distribution for the 4+ states) moves simulated chr21 reads from 0.04% to 2.54% inserted bases and from a fixed 1.00 bp to 2.83 bp mean insertion length, and raises the total error rate from 6.5% to 8.8% against a real 14.5%. As a single-variable ablation on *E. coli*, the encoding improves the reads-level composite from 83.5 to 87.6, driven by a 16-point gain in the k-mer sub-score. The residual gap to real data is concentrated in deletion run length and in read segments that alignment-based training cannot observe.

Reads-level fidelity across simulators

We compared GCerrHMM with NanoSim, PBSim3 (sample and errHMM modes) and badread [12] on the Level-1 reads-level panel (*E. coli*, *S. cerevisiae*, *H. sapiens* chr21; Fig. 5): read-length distribution, quality-value histogram, GC-content distribution and k-mer frequency correlation, summarised by a composite score with locked weights (Methods). Two of the four composite terms (read length and QV) are matched by construction because every run receives the same real-read profile; they verify profile application rather than simulator skill. The composite therefore functions as a descriptive summary of marginal distributions, and no GC-conditioning claim is drawn from it (see below).

On the composite, PBSim3-sample leads on *E. coli* (92.8) and chr21 (89.3), with GCerrHMM competitive on all three species (87.4 / 86.7 / 85.5; essentially tied with NanoSim on *S. cerevisiae*, 86.7 vs 86.8). PBSim3-errhmm is lowest (72.4 / 65.8 / 56.1), dragged down by its read-length term (16.0 on chr21) and QV term (45-48), the latter expected because its placeholder Q0 quality strings cannot reproduce a real QV histogram. On the k-mer term NanoSim leads on *E. coli* (91.1 vs 82.9 for GCerrHMM), while the GC-content term is high for all tools (96-99) except NanoSim on chr21 (64.8). As a single-variable ablation, the length-aware indel encoding improves GCerrHMM's composite on *E. coli* from 83.5 to 87.6, driven by a 16-point gain in the k-mer sub-score.

The HiFi cross-platform check completes end to end (composite 64-71 across the three arms, with wider intervals on the k-mer term; Fig. 5, right). We note that the public HiFi *E. coli* FASTQ

used here has an encoding-limited quality distribution (Phred-33 mean 80.28), so absolute QV agreement from that source is reported but not treated as a typical benchmark.

Alignment-level fidelity

Simulated reads were aligned with the same minimap2 and samtools [13, 14] pipeline and settings as the corresponding real data (Fig. 6). On human chr21, GCerrHMM achieves the highest mapping rate (88.7%; range across tools 85.8-88.7) and the second-highest mean MAPQ (34.1, vs 34.3 for PBSim3-sample); read-level base identity is led by PBSim3-sample (96.6%), followed by badread (94.8%), GCerrHMM (91.4%) and NanoSim (90.2%) - the ordering that the coverage-matched delta-to-real panel below converts into a distance ranking against the real anchor.

See Table 2.

Table 2. Alignment metrics for the external panel (mean; n=1 for each external tool-species row).

Species	Tool	Mapping rate (%)	Mean MAPQ	Base identity (%)	n
E. coli	GCerrHMM	100.00	59.02	99.26	1
E. coli	NanoSim	96.22	56.86	99.07	1
E. coli	PBSim3-sample	100.00	58.93	99.30	1
E. coli	PBSim3-errhmm	99.09	58.42	86.05	1
E. coli	badread	96.34	56.86	94.70	1
S. cerevisiae	GCerrHMM	100.00	52.21	98.97	1
S. cerevisiae	NanoSim	95.08	50.04	98.50	1
S. cerevisiae	PBSim3-sample	99.84	51.79	98.91	1
S. cerevisiae	PBSim3-errhmm	98.98	56.07	86.05	1
S. cerevisiae	badread	81.51	42.12	94.61	1
H. sapiens chr21	GCerrHMM	88.70	34.15	91.44	1
H. sapiens chr21	NanoSim	87.20	32.02	90.20	1
H. sapiens chr21	PBSim3-sample	88.63	34.34	96.57	1
H. sapiens chr21	PBSim3-errhmm	86.73	32.36	86.08	1
H. sapiens chr21	badread	85.79	31.81	94.78	1

n = one matched run per tool-species row. PBSim3-errhmm uses the same errhmm mode across species; its similar identity values are measured, not imputed, and each row comes from an independent run.

Variant calling: a cross-tool comparison without a real-data anchor

The variant layer scores every simulator against the same synthetic truth set (snp_rate = 0.001, indel_rate = 0.0001, seed 42), injected into the donor reference from which all reads are generated (Fig. 7). PBSim3-errhmm is omitted from this layer because its Q0 placeholder quality

strings make the standard caller non-evaluable; *H. sapiens* chr21 appears in the combined-F1 panel only, because PBSim3-sample reports no SNP/indel split. On chr21 combined SNP+indel F1, PBSim3-sample leads (94.9), with GCerrHMM (93.7) and badread (93.3) close behind and NanoSim markedly lower (65.6). On *E. coli* and *S. cerevisiae*, where the SNP/indel split is available for all plotted tools, combined F1 is 97.4-99.7 for all tools, and the SV spike-in panel separates no tool (69.0-78.0).

This layer has no real-data endpoint by design: real HG002 reads carry HG002's own variation and score $F1 = 0.0003$ against the synthetic truth, which is the expected result, not a pipeline failure. The variant layer is therefore excluded from the delta-to-real ranking below, and re-anchoring it to a published benchmark truth set (GIAB HG002 high-confidence calls) is stated as future work.

Assembly and phasing: delta-to-real on anchored layers

Two downstream layers carry a real-data anchor measured through the identical pipeline: alignment and assembly. On the assembly panel (Flye [15] for ONT, hifiasm for HiFi; PBSim3-errhmm omitted because no assembly run exists), assembly metrics were compared against the real 30x ONT anchor. The assembly and phasing summaries are shown in Fig. 8.

On the coverage-matched chr21 30x delta-to-real ranking, GCerrHMM places second on alignment identity (R3; delta 0.0489) and best on the assembly Borda distance (R5, 1.5 vs 2.0 for badread).

Table 3. Coverage-matched chr21 30x delta-to-real panel. R4 is excluded because its truth set is synthetic.

Tool	R3 identity (distance)	R5 identity	R5 N50 (Mb)	R5 Borda	Rank R3	Rank R5
GCerrHMM	0.9144 (0.0489)	0.9086	25.04	1.50	2.0	1.0
NanoSim	0.9020 (0.0365)	0.9166	10.04	3.00	1.0	3.0
PBSim3-sample	0.9657 (0.1002)	0.9460	21.48	3.50	4.0	4.0
badread	0.9478 (0.0823)	0.9212	33.74	2.00	3.0	2.0

R3 is the absolute difference in base identity from the real 30x ONT anchor. R5 Borda is the mean of the rank by absolute identity difference and the rank by $|\log_2(N50_{\text{tool}}/N50_{\text{real}})|$. R4 is excluded because its truth set is synthetic and has no real anchor.

For phasing, real HG002 HiFi reads processed with WhatsHap [16] achieved a phased fraction of 0.9655 across 179 blocks with block N50 of 177,187 bp. Switch error is not reported because no phased truth set is bundled with the panel. The spike-in donor designs are haploid and carry no heterozygous sites, so no simulated-read phasing arm is reported. The phasing panel is therefore a real-data quality check, not a cross-tool comparison.

See Table 4.

Table 4. Assembly N50 and reference identity for the simulated ONT panel and the real 30x ONT anchor (mean; n=1 per row).

Species	Tool	Assembler	Contigs	N50 (Mb)	Reference identity (%)	n
E. coli	GCerrHMM	Flye	3	3.68	98.35	1
E. coli	NanoSim	Flye	4	3.09	98.31	1
E. coli	PBSim3-sample	Flye	5	1.44	98.31	1
E. coli	badread	Flye	1	4.64	97.68	1
S. cerevisiae	GCerrHMM	Flye	45	0.63	97.70	1
S. cerevisiae	NanoSim	Flye	49	0.54	98.07	1
S. cerevisiae	PBSim3-sample	Flye	49	0.50	97.66	1
S. cerevisiae	badread	Flye	45	0.60	96.96	1
H. sapiens chr21	GCerrHMM	Flye	40	25.04	90.86	1
H. sapiens chr21	NanoSim	Flye	51	10.04	91.66	1
H. sapiens chr21	PBSim3-sample	Flye	36	21.48	94.60	1
H. sapiens chr21	badread	Flye	43	33.74	92.12	1
H. sapiens chr21	Real 30x ONT anchor	Flye	48	33.50	79.77	1

The real 30x ONT row is the anchor used for the delta-to-real comparison in Table 3; simulated rows are shown for context. Reference identity is matched bases divided by aligned bases from an asm5 PAF alignment of the unpolished Flye contigs against the reference. It is not a polished consensus identity and is the same metric used for every simulated row.

GC-bin ablation and the pre-registered decision

Two instruments, and why they answer different questions

The composite aggregates read-length, QV, GC-content and k-mer terms; it is designed for marginal distributional similarity and is reported as a descriptive summary. The GC claim is evaluated on a GC-stratified error curve - per-window GC bins (100 bp, equal width) and the error rate of aligned bases within each bin - compared between real and simulated reads by

Pearson correlation and mean absolute deviation, with explicit intervals (Methods). A one-bin model cannot generate a GC-stratified profile; its finite-sample correlation therefore reflects alignment between the global error level and the real curve, not GC-conditional capability. The pre-registered decision compares the one-bin control against the GC-aware model across bins and species rather than testing whether a near-zero one-bin correlation is a separate hypothesis.

Composite separability and stratified evidence

On the profile-matched bin ablation (1, 5, 10, 20 bins; 3 seeds each), the composite score shows no separable difference across bin settings in either ablation species (*A. thaliana* 92.6-96.6; *E. coli* 84.8-87.6, overlapping intervals; Fig. 9a). Training coverage and state-space simplification likewise leave the composite flat (Fig. 9b, c).

On the six-species panel, the GC-aware model exceeded its one-bin control in both replicates for *E. coli* (0.458/0.589 vs. -0.738/0.505), *S. cerevisiae* (0.471/0.535 vs. 0.230/0.174), *M. musculus* chr19 (0.922/0.794 vs. 0.460/0.650) and *H. sapiens* chr21 (0.917/0.861 vs. -0.579/-0.091). *A. thaliana* and *D. melanogaster* were mixed. The strongest directional signal came from *H. sapiens* chr21; the pre-registered double-species gate defines the scope of the cross-species claim, and the human panel remains a two-replicate supporting analysis. In the 10x cross-tool panel, the comparison provides a conservative reference point; the GC-stratified six-species panel is reported in Table 7.

Table 5. GC-bin ablation scored by curve correlation and MAD.

Species	GC bins	Curve r, mean (SD)	MAD pp, mean (SD)
<i>E. coli</i>	1	0.077 (0.334)	0.255 (0.006)
<i>E. coli</i>	5	-0.080 (0.571)	0.252 (0.004)
<i>E. coli</i>	10	0.233 (0.504)	0.258 (0.003)
<i>E. coli</i>	20	0.576 (0.184)	0.266 (0.007)
<i>A. thaliana</i>	1	0.779 (0.062)	4.204 (0.010)
<i>A. thaliana</i>	5	0.960 (0.011)	4.212 (0.039)
<i>A. thaliana</i>	10	0.970 (0.002)	4.475 (0.015)
<i>A. thaliana</i>	20	0.981 (0.004)	4.398 (0.039)

This profile-matched three-seed bin ablation (*E. coli* and *A. thaliana*) is a different experiment from the six-species, two-replicate route panel reported in Table 7.

The noise floor, quantified

The 95% between-window bootstrap interval on the GC-curve correlation was 1.01 at 10x, 0.96 at 20x and 0.96 at 30x. It does not shrink with sequencing depth and is a structural uncertainty of the window-level curve, not a power statement. In the separate depth grid, the read-sampling SD was 0.176 at 5x (6 disjoint subsets), 0.194 at 10x (3 subsets) and 0.009 at 15x (2 halves). The 15x estimate is therefore suggestive rather than precise; it indicates that an effect larger than about 0.05 correlation units would have been detectable above 15x, while the between-window interval remains wide.

Table 6. Depth dependence of the read-sampling component.

Depth	Disjoint subsets	Mean r	SD across subsets
5x	6	0.047	0.176
10x	3	0.139	0.194
15x	2	0.008	0.009

The 5x, 10x and 15x values come from 6, 3 and 2 disjoint subsets, respectively. The 15x estimate is based on two halves and is reported as suggestive. The between-window interval is a separate, depth-independent uncertainty and is not a power statement. The subset count and per-subset coverage differ across depths, so the SD estimates are not directly comparable; the 10x estimate is based on three subsets and is not expected to be monotonic relative to the six-subset 5x estimate.

Six-species panel: distributional and GC-stratified results

Across the six-species route panel, the k-mer statistic is not a GC-stratified instrument: 11 of 12 route/replicate comparisons had a difference interval containing zero, and the single interval excluding zero favoured the 1-bin control without reproducing. That k-mer panel is therefore reported as not discriminable for distributional composition. The GC-stratified curve comparison is reported separately in Table 7: the GC-aware model is directionally favoured in both replicates for four of six genomes, while *A. thaliana* and *D. melanogaster* are mixed. The pre-registered double-species gate defines the scope of the cross-species claim; we report the observed directional evidence and effect sizes.

Verified against the v2 six-species panel: 4/6 genomes favour GC-aware in both replicates; the exceptions are *A. thaliana* and *D. melanogaster*.

236 **Table 7. Six-species directional panel. Entries are Pearson r against the real GC-error curve; both replicates are shown.**

Species	B (1-bin) r1 / r2	C (GC-aware) r1 / r2	Direction
<i>E. coli</i>	-0.738 / 0.505	0.458 / 0.589	C > B, both
<i>S. cerevisiae</i>	0.230 / 0.174	0.471 / 0.535	C > B, both
<i>A. thaliana</i>	0.325 / 0.902	0.947 / 0.862	Mixed
<i>D. melanogaster</i>	-0.778 / -0.756	-0.882 / -0.570	Mixed
<i>M. musculus</i> chr19	0.460 / 0.650	0.922 / 0.794	C > B, both
<i>H. sapiens</i> chr21	-0.579 / -0.091	0.917 / 0.861	C > B, both

237 This is the six-species B/C route panel, not the profile-matched bin ablation in Table 5. *H. sapiens* chr21 shows
 238 the largest directional contrast, but it is a two-replicate supporting panel, not the pre-registered two-species
 239 decision gate.

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242 **Discussion**

243 We asked whether conditioning a long-read error model on local GC content improves
244 simulation fidelity and whether standard evaluation metrics can detect that improvement. We
245 found measurable GC-conditional error structure and directionally closer GC-error curves in four
246 of six genomes; the pre-registered gate defines the scope of the cross-species claim. The broader
247 lesson is that a fidelity score can test only the property it contains; evaluation instruments and
248 claims must be aligned.

249 **A testable boundary hypothesis**

250 The six-genome panel spans GC-uniform and strongly GC-heterogeneous genomes and
251 provides directional support for a boundary hypothesis: the strongest positive case was *H.*
252 *sapiens* chr21, while *A. thaliana* and *D. melanogaster* were mixed. The pre-registered directional
253 gate defines the scope of the cross-species claim, and the human panel remains a two-replicate
254 supporting analysis. A future pre-registered panel with more genomes and independent read sets
255 should test whether the model-implied GC-error amplitude predicts when GC-aware training is
256 beneficial.

257 **Instrument-claim alignment as a reporting principle**

258 The composite score was not wrong; it was asked the wrong question. Because none of its
259 terms is a function of local GC content, no configuration of a GC-aware model can move it
260 through the GC mechanism - and indeed the bin ablation leaves it flat while the stratified
261 instrument moves. We therefore report the composite as a descriptive summary of marginal
262 distributions and reserve the GC-stratified error curve, with explicit intervals, for the GC claim.
263 The same discipline resolved an apparent contradiction in our depth analysis: read-sampling
264 spread shrinks with depth while the between-window bootstrap does not, and only the former is
265 a power statement. Conflating the two produces contradictory conclusions from the same data.

Four measurement problems that produced plausible but wrong numbers

Four measurement problems were found and fixed during development, each of which had produced plausible but wrong numbers: a MAPQ threshold that removed the reads carrying most of the error signal; a single-insertion-state encoding that capped every insertion at 1 bp; a missing read-length profile that silently fell back to a platform default; and a k-mer sub-score whose replicate-to-replicate spread exceeded the between-route differences it was used to rank. We report them because they determine how any simulator benchmark in this space should be read, and because the pipeline now fails loudly instead of degrading quietly.

Relation to existing long-read simulators

NanoSim, PBSim3 and NPBSS established the value of empirical or HMM-based long-read simulation [1-3]. GCerrHMM keeps the same alignment-derived, fail-loud design but makes local GC content an explicit state variable and evaluates that variable with a pre-registered stratified instrument.

Table 8. Relationship to representative long-read simulators.

Tool	Primary model	Local context	Evaluation emphasis	Relationship to this study
NanoSim [1]	Alignment-derived error profiles and Markov chain	Read/alignment profile	Read composition and alignment	External baseline; no explicit GC-conditioned transition state
PBSim3 [2]	HMM error model	Platform/error profile	Quality and error profiles	External baseline; multi-platform capability
NPBSS [3]	Empirical error and quality model	Read/error profile	Error-QV and assembly	Methodological comparator for empirical simulation
GCerrHMM	GC-aware error HMM with run-length indel states	100-bp reference window and GC bin	GC-curve fidelity and delta-to-real	Models the GC-dependent error structure explicitly

Limitations

First, the GC-conditional effect is reported as directional evidence with effect sizes across 10-30x; the pre-registered gate defines the scope of the cross-species claim, and the measured depth budget is reported in the Abstract. Second, the Level-1 composite measures distributional similarity only; conclusions about GC conditioning rest on the stratified-error instrument and the controlled bin ablation. Third, the k-mer sub-score has little discriminating power for genomes with a small shared k-mer intersection (*D. melanogaster* $r = 0.12-0.35$ vs. *E. coli* $0.86-0.94$) and a replicate spread of up to 16.7 points. Fourth, the variant-calling layer uses a synthetic truth set

and has no real-data anchor; re-anchoring it to GIAB HG002 high-confidence calls is future work, as is the 6.62x assembly condition where real reads do not assemble. Fifth, phasing switch error could not be reported for lack of a bundled phased truth. Sixth, the public HiFi E. coli FASTQ used for the cross-platform check has an encoding-limited quality distribution, so HiFi QV agreement is indicative only.

Conclusions

GCerrHMM is a trainable GC-aware error model for long-read simulation, with a length-aware indel encoding, competitive compositional fidelity, and downstream consistency that ranks first or second among tested simulators on real-anchored layers. Its GC-conditional effect is measurable in structure; we report effect sizes, the separate sampling and structural uncertainties, and the depth budget required for a definitive cross-species test. The study demonstrates that simulator benchmarking requires instrument-claim alignment: a fidelity score can test only the property it contains. The fail-loud harness, instrument audit and pre-registered decision framework are released with the model.

Methods

Overview of the simulation framework

The framework has four stages: (i) alignment of real long reads to a reference; (ii) training of the GC-aware error HMM from the alignments; (iii) read generation from a target reference with platform-specific profiles; and (iv) two-level evaluation - Level-1 distributional fidelity and Level-2 downstream task consistency - against both real data and competing simulators. All routes and tools share mapper, caller and assembler settings within each evaluation layer, and every simulated dataset that feeds a table or figure passes a fail-loud gate (below).

Software: minimap2 2.31-r1302 (map-ont for ONT; map-hifi for HiFi), samtools 1.23.1, bcftools 1.24, Flye 2.9.6-b1802, hifiasm 0.25.0, WhatsHap 2.8, Python 3.11.16, pysam 0.24.1, NumPy 2.4.6, SciPy 1.17.1, Matplotlib 3.11.2 and Biopython 1.88.

The GC-aware errHMM

Model definition

For each 100-bp reference window, local GC content is assigned to one of K equal-width bins. The model estimates $P(\text{state at } i+1 \mid \text{state at } i, \text{GC bin})$ from the CIGAR path together with a nucleotide substitution matrix. Conditioning therefore applies to the state-transition matrix; the substitution matrix is global rather than GC-specific. The default state space is M, S, I1, I2, I3, I4+, D1, D2, D3, D4+; a simplified state space merges the indel states into single I and D states and serves as an ablation control. The one-bin control ($K = 1$) is trained on the same aligned reads with the same MAPQ threshold, GC window and read budget.

Length-aware indel encoding

Indel run length is carried in the state name, for insertions and deletions alike. A run of one, two or three bases enters I1/I2/I3 or D1/D2/D3; longer runs share I4+/D4+, whose length distribution is learned separately and is heavy-tailed. Among chr21 insertion runs of 4 bp or more,

35% are exactly 4 bp, 52% are 5-9 bp, 10% are 10-49 bp and 3.5% are at least 50 bp, with a mean of about 241 bp in that final tail. With a single I state the I-to-I transition is never observed and the model emits every insertion as exactly 1 bp; the quantitative consequences are reported in Results.

Parameter estimation from real data

Training parses primary alignments from a real long-read BAM with pysam, labels each reference position as match, substitution, insertion or deletion from the CIGAR and MD tags, and counts transitions per GC bin. Transition probabilities use hierarchical smoothing: pooled transition counts form a global prior, and each GC bin is smoothed toward that prior. The process is fully observed, so estimation is counting-based maximum likelihood rather than Baum-Welch. The trained model is stored as JSON.

Panel models used the default minimum MAPQ of 20, except for the human chr21 30x model. Chr21 was the only species whose read pool fell below the 20,000-read cap: 58% of its 8,189 primary reads mapped below MAPQ 20, so applying the default retained 3,438 reads and biased the learned error spectrum (event rate 0.0459 at MAPQ 20 vs 0.0713 at MAPQ 0). The chr21 model therefore uses minimum MAPQ 0; the other species retained MAPQ 20 because their larger pools were capped at 20,000 reads and the threshold changed the retained event rate by less than 0.1 percentage points. This is a platform-specific correction to an alignment-depth artefact, not a different model class. Hierarchical smoothing used prior strength 10.0 with pseudocount 1.0.

Read generation and implementation

Reads are sampled from a target reference using platform-specific length profiles or a real-read profile. At each reference position the simulator reads the GC bin and samples the next state from the corresponding transition row; substitutions use the learned nucleotide matrix, and indels, read lengths and quality values are generated from the route-specific model and profile. The generator samples read layouts in the parent process, assigns a deterministic seed to each read,

and generates fixed-size chunks in worker processes (capped at 16), so generated FASTQ is invariant to worker count.

Reference genomes and real sequencing data

Six reference panels are used in the frozen analysis: *E. coli*, *S. cerevisiae*, *A. thaliana*, *D. melanogaster*, human chr21 (HG002) and mouse chr19, spanning GC-uniform to GC-heterogeneous genomes. ONT accessions are recorded in the repository configuration; real HiFi sources are recorded separately with accessions, file sizes and checksums.

Table 1. Analyzed species panels and data sources.

Species	RefSeq assembly	Size (Mb)	GC (%)	GC heterogeneity (SD)	ONT run
<i>E. coli</i>	GCF_000005845.2	4.6	50.8	4.8	SRR39619343
<i>S. cerevisiae</i>	GCF_000146045.2	12.2	38.1	3.8	SRR39791672
<i>A. thaliana</i>	GCF_000001735.2	119.7	36.1	5.5	ERR5716408
<i>D. melanogaster</i>	GCF_000001215.4	143.7	42.0	6.3	SRR22071686
<i>M. musculus</i> chr19	GCF_000001635.27	61.4	42.7	5.7	SRR14685232
<i>H. sapiens</i> chr21	GCF_000001405.40	51.5	41.2	8.2	ERR13491992

The species table is generated from config/species_config_v2.json and config/data_sources.json; Fig. 4c lists reference and ONT run accessions. Training uses the real alignment panels listed there, whereas evaluation uses matched simulated FASTQ at the coverage stated in each Results table; no simulated reads are used to train the reported model.

Simulation design: routes and the cross-tool panel

Three within-framework routes isolate the scientific questions: the sample route (empirical error sampling) as the real-sampling baseline, the one-bin route ($K = 1$) as the GC-unaware methodological control, and the GC-aware route (full GC-conditioned model). Replicates differ only in random seed. The cross-tool panel compares GCerrHMM with NanoSim, PBSim3 (sample and errHMM modes) and badread; all tools receive the same real-read profile and target coverage, and all routes and tools use the same truth-variant reference so that truth variants are present in every simulated read set.

The external cross-tool panel uses one simulated replicate per tool and species; the within-

framework B/C panel uses two replicates, and the GC-bin ablation uses three seeds at 10x. PBSim3-errhmm is retained in reads and alignment panels because those metrics do not require quality values, omitted from variant panels because Q0 placeholder qualities make the quality-filtered caller non-evaluable, and omitted from assembly because no assembly was run. PBSim3-sample is included in all applicable panels; chr21 has no SNP/indel split, so it appears only in the combined-F1 panel.

Level-1 evaluation: distributional fidelity

Level-1 compares read-length, QV, GC and k-mer distributions between real and simulated FASTQ. The composite uses locked weights (read_length 0.15, QV 0.20, GC 0.25, k-mer 0.15), renormalised over available terms; the error-rate term is disabled for the reported panel. Two terms (read length, QV) are matched by construction via the shared real-read profile and primarily verify profile application. None of the four terms is a function of local GC content; the composite is therefore reported as a descriptive summary and is not used as evidence for the GC claim.

The GC claim is tested with an aligned instrument: per-window GC bins (100 bp, equal width, the trainer's convention) and the error rate of aligned bases within each bin from the CIGAR with soft-clipped bases excluded. Real and simulated sets are aligned with the same mapper; the simulated per-bin error profile is compared with the real one by Pearson and Spearman correlation and mean absolute deviation. A one-bin model cannot generate a GC-stratified error profile, so its finite-sample correlation across GC bins reflects alignment between the global error level and the real curve, not GC-conditional fidelity; the pre-registered decision compares the one-bin control against the GC-aware model rather than testing whether the one-bin correlation is identically zero. The k-mer term correlates frequency vectors over the intersection of per-side top-10,000 k-mer maps ($k = 21$, Pearson); the sample was increased from 250 to 1000 reads per side with a paired bootstrap over reads after the 250-read estimator showed a replicate spread of up to 16.7 points.

Level-2 evaluation: downstream task consistency

Alignment: simulated and real reads are aligned with the same mapper and settings; mapping rate, MAPQ, CIGAR-derived error rates and read-level identity are summarised. Variant calling: normalised VCFs are compared against the truth VCF with the same caller and filters for all routes and tools; because the truth set is synthetic and injected into the shared donor reference, this layer is a cross-tool comparison and is excluded from delta-to-real. Assembly: hifiasm for HiFi reads, Flye for ONT reads. Phasing: whatshap against a phased truth VCF where available; for the HG002 panel no phased truth is bundled, so phased fraction and block statistics are reported without switch error. Delta-to-real rankings are computed only on layers with a real-data anchor processed through the identical pipeline (alignment, assembly).

Ablation study

Two ablation axes were pre-registered. GC-bin granularity: K in $\{1, 5, 10, 20\}$ with profile-matched training and three seeds per setting, evaluated on both the composite and the GC-stratified instrument; the pre-registered gate defines the scope of the cross-species significance claim. Training coverage: matched panels across a coverage grid, evaluated on the composite. The observed effects are reported with effect sizes and the measurement budget.

Statistical analysis and reproducibility

Replicates differ only in random seed. Paired bootstrap intervals resample read indices once and reuse them for both routes. Uncertainty in the GC-fidelity instrument is decomposed into a read-sampling component (disjoint subsamples of a pooled 30x simulation) and a between-window bootstrap component; only the former is read as a power statement. Every simulation feeding a table or figure passes a fail-loud gate: an entry check refuses to start without a read-length and quality profile, and an exit check recomputes produced read count, total bases and mean read length against the request (coverage x reference length, profile mean) with 10% tolerance; violations abort the run before downstream consumption. Expected-unattended jobs are supervised and restarted on silent failure. The repository provides reproduce.sh with --gc-

427 demo (retraining and matched simulation) and --framework-figures (evaluation tables and figure
428 suite).

429 Repository URL and Zenodo DOI are author-controlled release fields and must be supplied
430 after deposition. Wall-clock runtimes are machine-dependent and are recorded in the
431 reproducibility audit accompanying the release.

432

433 **Declarations**434 **Ethics approval and consent to participate: Not applicable.**435 **Consent for publication: Not applicable.**436 **Availability of data and materials: [Repository URL and Zenodo DOI to be supplied after**
437 **deposition]. Public accessions are listed in Table 1.**438 **Competing interests: [AUTHOR INPUT REQUIRED].**439 **Funding: [AUTHOR INPUT REQUIRED: grant numbers or 'none'].**440 **Authors' contributions: [AUTHOR INPUT REQUIRED: CRediT roles].**441 **Acknowledgements: [AUTHOR INPUT REQUIRED or 'none'].**

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Figure legends

Figure 1. Local GC context across the six-genome panel and an illustrative real-versus-simulated GC-error curve for *E. coli*. Panel a uses the six reference genomes; panel b uses the project GC-error curve. No inferential error bars are shown.

Figure 2. Recent long-read simulation and benchmark landscape. Panel a shows selected publications from 2017-2026. Panel b is a qualitative summary of primary model features based on the cited papers; yes, partial and - indicate whether the feature is a primary design element, a supporting component or not a primary model axis, and are not performance scores. The matrix is a qualitative design summary, not a performance score. No inferential error bars are shown.

Figure 3. Context dependence and evaluation layers. Panel a shows the observed deletion rate as a function of homopolymer length in the project's real *E. coli* ONT panel; panel b is a qualitative synthesis of the modelling and evaluation layers addressed by the cited recent work. The layers are not performance scores, and the panel is intended to motivate metric-aligned evaluation rather than rank tools. Panel a uses the frozen real *E. coli* ONT homopolymer table; panel b is a qualitative synthesis of the cited literature. No inferential error bars are shown.

Figure 4. (a) Dual-platform workflow from aligned ONT or PacBio HiFi BAM through CIGAR-derived error-state learning, GC-conditioned transition modeling, matched simulation, and four-layer evaluation. (b) Model-implied error state varies across GC bins for representative genomes. (c) Species panel with the analyzed reference accession, genome size, reference GC content, 1-kb GC heterogeneity, and real ONT run accession. n = six reference panels; no error bars (workflow and species-source table).

Figure 5. Matched-data comparison of GCerrHMM, NanoSim, PBSim3-sample, PBSim3-errhmm, and badread using read-length, QV, GC, and k-mer similarity. The current measured facet is ONT; the HiFi facet is emitted only after the same training, simulation, and matched-sampling evaluation has completed. All plotted read-level cells are measured; PBSim3-errhmm is retained because these read statistics do not depend on its Q0 quality

strings. n = one matched simulation per species and tool; heatmap values are point estimates and no error bars are shown.

Figure 6. Mapping rate, mean MAPQ, and base-level identity after a common minimap2/samtools alignment pipeline. All plotted cells are measured. PBSim3-errhmm is retained here because its mapping statistics are valid, even though its Q0 placeholder qualities make standard variant calling non-evaluable. The *H. sapiens* chr21 row uses the coverage-matched 30x panel. n = one matched alignment run per species and tool; heatmap values are point estimates and no error bars are shown.

Figure 7. Combined SNP/indel F1 for *E. coli*, *S. cerevisiae*, and *H. sapiens* chr21 against the same truth VCF, plus SNP-only, indel-only, and SV spike-in F1 for the two species where every plotted tool has that measurement. The *H. sapiens* chr21 row uses the coverage-matched 30x panel; this layer is reported as a cross-tool comparison and is excluded from delta-to-real. PBSim3-errhmm is omitted because its Q0 placeholder qualities make the standard caller non-evaluable. *H. sapiens* chr21 is shown in the combined-F1 panel only because PBSim3-sample has no SNP/indel split. n = one matched caller run per species and tool; SV values pool the four spike-in designs and confidence intervals are reported in the accompanying table.

Figure 8. Assembly N50 and reference identity for the measured panel, together with the phasing status. HiFi is assembled with hifiasm, ONT with Flye, and diploid phasing is evaluated with whatshap against a phased truth VCF. Historical raven values are not relabeled as hifiasm or Flye results, and PBSim3-errhmm is omitted because no assembly run exists for it. The *H. sapiens* chr21 row uses the coverage-matched 30x panel; N50 colors span 0-40 Mb so that the real-anchor and simulated ranges remain visible. n = one Flye assembly per species and tool, plus one real 30x ONT anchor; no error bars are shown.

Figure 9. (a) Profile-matched GC-bin ablation for *E. coli* and *A. thaliana* across three seeds. (b) Training-coverage ablation from 1x to 30x. (c) Full versus simplified state-space models.

527 **Error bars are mean $\pm$ SD. n = three seeds per condition; error bars are mean $\pm$ SD.**

528 **Additional files**

529 Additional file 1. File name: Additional_file_1_species_panel.xlsx. File format: XLSX. Title:

530 Species panel and source accessions. Description: reference accessions, genome size, GC

531 content, GC heterogeneity and ONT run accessions.

532 Additional file 2. File name: Additional_file_2_level1_sub_scores.xlsx. File format: XLSX.

533 Title: Full per-species Level-1 sub-scores and HiFi cross-platform scores. Description:

534 replicate-level read-length, QV, GC and k-mer scores.

535 Additional file 3. File name: Additional_file_3_gc_bin_ablation.xlsx. File format: XLSX.

536 Title: GC-bin ablation full table. Description: curve r and MAD by species, bin setting and

537 seed.

538 Additional file 4. File name: Additional_file_4_delta_to_real.xlsx. File format: XLSX. Title:

539 Delta-to-real per-tool table. Description: anchored-layer metrics and ranks.

540 Additional file 5. File name: Additional_file_5_instrument_audit.pdf. File format:

541 PDF/DOCX. Title: Instrument audit. Description: four measurement problems, symptoms and

542 fixes.

543 Additional file 6. File name: Additional_file_6_uncertainty_layers.pdf. File format:

544 PDF/DOCX. Title: Two layers of uncertainty. Description: read-sampling and between-

545 window components of the GC-fidelity instrument.

546 Additional file 7. File name: Additional_file_7_hifi_crossplatform_note.pdf. File format:

547 PDF/DOCX. Title: HiFi cross-platform note. Description: cross-platform chain and the

548 encoding-limited QV caveat.